

brackets give the size of the filter, the default value of which is 3x3. The final parameter is the standard deviation. If any value is not specified, a default value of 0.5 is assumed. In the command given above, since the standard deviation parameter is not specified, default value of 0.5 is taken.

Fig. 8 shows a Gaussian filtered image with standard deviation value taken as 0.5.

When the standard deviation is specified to be 5, as shown in the command (`G1 = fspecial('gaussian',[11,11],5);`), the resultant image is as shown in Fig. 9.

Linear high-pass filters

Linear HPFs can be implemented using 2D convolution masks with positive and negative coefficients, which correspond to a digital approximation of the Laplacian—a simple, isotropic (rotation-invariant) second-order derivative that is capable of responding to intensity transitions in any direction.

An alternative digital implementation of the Laplacian takes into account all eight neighbours of the reference pixel in the input image and can be implemented by the convolution mask given below:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Directional difference filters.

Directional difference filters are similar to the Laplacian high-frequency filter but emphasise edges in a specific direction. There are four representative masks that can be used to implement the emboss effect:

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

Linear high-pass filters include Prewitt and Sobel operators. These are also referred to as first-order edge-detection methods. High-pass filters are used for edge detection



Fig. 9: Gaussian filtered image with standard deviation specified as 5



Fig. 10: Image with edges extracted using the Prewitt operator

and edge enhancement operations.

The IPT function edge has options for both Prewitt and Sobel operators. Edge detection using Prewitt and Sobel operators can also be achieved by using `imfilter` command with the corresponding 3x3 masks (which can be created using `fspecial` command).

Following code extracts edges in the image using Prewitt operator:

```
>> A = imread('Penguins_grey.jpg');
>>Prewitt_A=edge(A,'Prewitt');
>>imshow(Prewitt_A)
```

Fig. 10 shows the resultant image.

Let us now add Gaussian noise to the `Penguins_grey.jpg` image. We then process it using the Prewitt edge detector:

```
>> A = imread('Penguins_grey.jpg');
>>A_noise=imnoise(A,'gaussian');
>>Prewitt_A_noise=edge(A_noise,
'Prewitt');
>>imshow(A_noise)
```

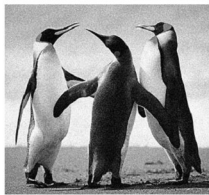


Fig. 11: Image with the added Gaussian noise



Fig. 12: Image filtered with Prewitt edge detector

```
>>imshow(Prewitt_A_noise)
```

Fig. 11 shows the image with the added Gaussian noise and Fig. 12 shows the image filtered with Prewitt edge detector.

Following code shows the use of Sobel operator for edge detection:

```
>> A = imread('Penguins_grey.jpg');
>>Sobel_A=edge(A,'sobel');
>>imshow(Sobel_A)
```

Fig. 13 shows the resultant image.

Let us now add Gaussian noise to the `Penguins_grey.jpg` image. We then process it using the Sobel edge detector:

```
>> A = imread('Penguins_grey.jpg');
>>A_noise=imnoise(A,'gaussian');
>>Sobel_A=edge(A,'sobel');
>>imshow(Sobel_A)
```

Fig. 14 shows the resultant image.

Non-linear filters

Non-linear filters work at a neighbourhood level, but do not process